Machine Learning — Exercise 0

Dataset Description | Group: [GROUP NUMBER] | Members: [NAME 1], [NAME 2], [NAME 3]

1. Dataset selection and overview

We selected two classification datasets that differ in sample size, attribute types, missingness, and target distribution. Credit Card Fraud Classification is useful for studying a large dataset with a rare positive class; Adult ASD Screening is smaller and combines binary questionnaire responses with demographic categories. This contrast lets us describe how data representation and data quality influence classification preparation and evaluation. Replace this rationale with the group's actual selection reasoning if it differs.

Characteristic	Credit Card Fraud Classification	Adult ASD Screening
Original size	28,480 samples; 31 attributes	704 samples; 21 attributes
Inputs and target	30 inputs; target: class	20 inputs; target: Class/ASD
Types	Numerical features; nominal target	Binary scores, numerical fields, nominal fields
Target	otherwise 28,431 (99.83%); fraud 49 (0.17%)	NO 515 (73.15%); YES 189 (26.85%)
Missing cells	None detected	192 in age, ethnicity, relation

2. Measurement scales and attribute ranges

The fraud dataset has numerical time, amount, and v1–v28 fields, with a nominal categorical target. Amount is a ratio-scale quantity (a meaningful zero and ratios); time is numeric, but its precise measurement interpretation should follow the dataset documentation. The transformed v1–v28 fields are numerical inputs, but their original measurement scales cannot be assigned confidently from column names alone. Amount ranges from 0 to 8,360 (median 22.00; mean 86.76), and the observed distribution has a long right tail.

In the ASD dataset, A1_Score–A10_Score are binary indicators (0/1), result ranges from 0 to 10, and age is a ratio-scale quantity in principle. Gender, ethnicity, jundice, austim, contry_of_res, used_app_before, age_desc, relation, and Class/ASD are nominal categorical fields. No original field is confidently identified as ordinal from the inspected attributes. The result field appears to equal the sum of the ten binary scores; verify this from the data before treating it as a confirmed relationship.

Why scales matter: nominal categories have no inherent order and should not be treated as numeric distances; ordinal categories preserve order; interval measurements support meaningful differences but not meaningful ratios; ratio measurements have a meaningful zero and support ratios. These distinctions inform encoding, summaries, and model preprocessing.

3. Distributions and target imbalance

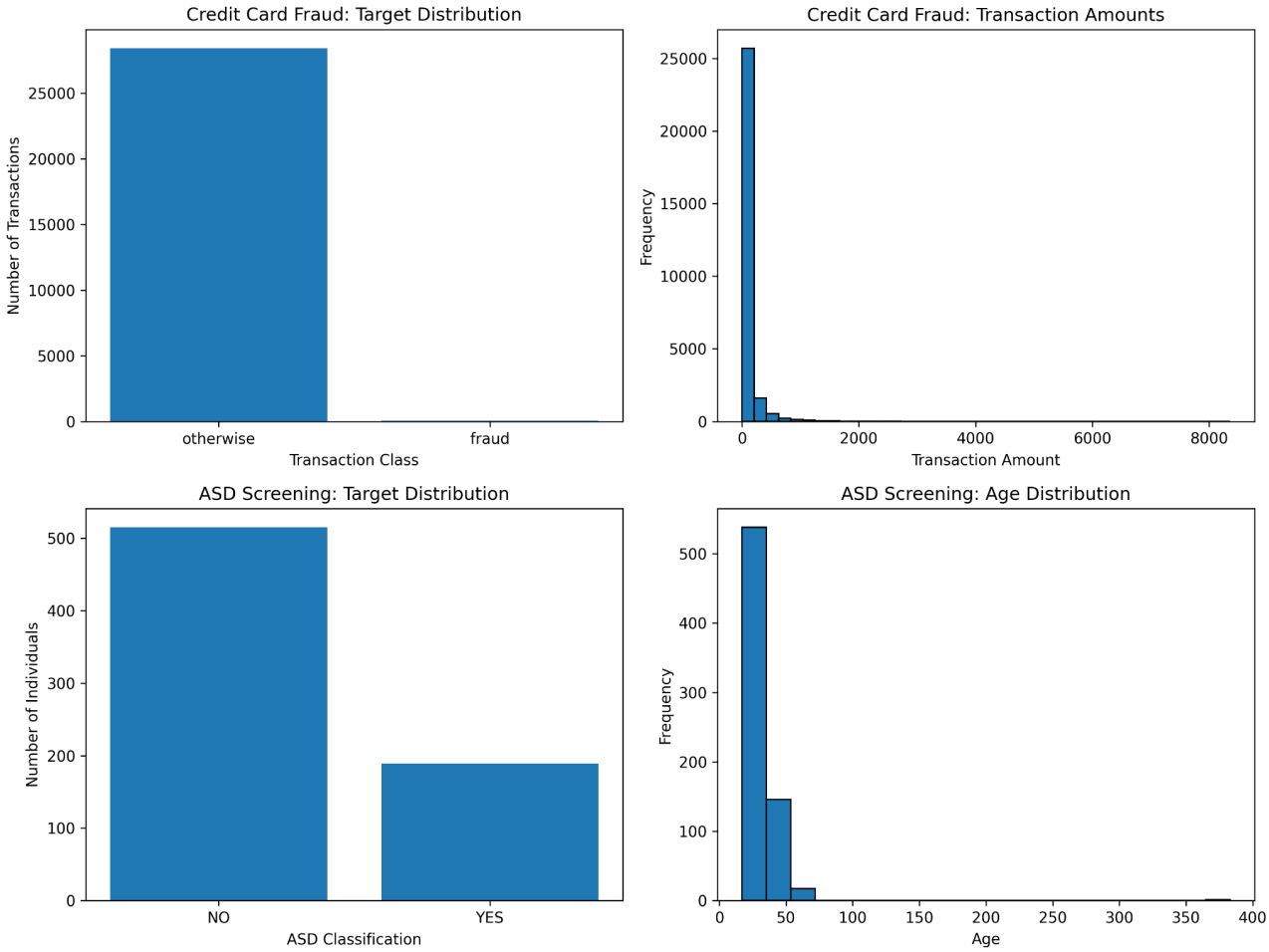


Figure 1. Target distributions and selected numerical distributions. The plotting script uses a log frequency axis for transaction amounts.

The fraud target is severely imbalanced: fraud represents 0.17% of records. A classifier predicting only the majority class could have high accuracy while detecting no fraud, so accuracy alone would be misleading. The ASD target is also imbalanced, with 26.85% YES observations, but less extremely.

4. Data quality and potential preprocessing

No missing values were detected in the original fraud attributes. In the ASD dataset, age has 2 missing values, ethnicity 95, and relation 95, totaling 192 missing cells in the original attributes. One age value is 383 (record index 52), an implausible human age; it is flagged rather than silently changed. The age_desc field has one observed category and may provide no useful variation. Some category strings appear to include quote characters; verify whether these are literal values or ARFF parsing artifacts before normalizing them. The result field appears to be the sum of A1_Score–A10_Score and should be checked programmatically. Potential later steps include documented missing-value handling, categorical encoding, and algorithm-appropriate scaling; fit transformations on training data only.

5. Group contributions

[NAME 1]: [DESCRIBE CONTRIBUTION]

[NAME 2]: [DESCRIBE CONTRIBUTION]

[NAME 3]: [DESCRIBE CONTRIBUTION]

Reproducibility

The accompanying ZIP contains original datasets, editable DOCX report source, scripts, requirements, and generated plots. Run scripts/plot_distributions.py and scripts/feature_analysis.py from the project root after installing requirements.txt. Original data files are not modified.